

SynapseLab

Synaptic Plasticity as Short-Term Memory

DataForge 2026 — Pathway Track Educational Blog

1. The Core Idea

Short-term memory does not have to mean keeping a continuously active representation. A different mechanism is to temporarily change the strength of connections between units. This is the central idea behind **synaptic plasticity as short-term memory**: recent activity leaves a temporary trace in synaptic state, and that trace can influence what happens next.

SynapseLab is built around one falsifiable teaching claim: **AI systems can use changing synaptic connections as temporary memory, but conflicting information can interfere with that memory**. The learner tests this claim by teaching the same input different associations and watching the memory strengths change.

2. Why This Matters for AI

Conventional sequence models often represent context through activations, recurrent state, or attention over a context window. Synaptic-memory approaches introduce another design choice: part of the information state is stored in connection strengths that change during processing. This separates fast neural activity from a slower, temporary state carried by connections.

Research on short-term synaptic plasticity supports this as a serious computational mechanism rather than only a biological metaphor. Kozachkov et al. (2022) compared recurrent neural networks with and without short-term synaptic plasticity on a working-memory task. Both could maintain memories, while the plasticity-based networks showed more brain-like activity and greater robustness to network degradation. [1]

A 2023 review by Lansner, Fiebig and Herman describes fast Hebbian plasticity as an alternative to purely persistent neural activity for working memory, motivated in part by evidence for activity-silent forms of memory. [2]

3. A Simple Mathematical View

Let w_{ij} represent the temporary strength of a connection from unit i to unit j . A simple educational Hebbian update can be written as:

$$w_{ij}(t+1) = w_{ij}(t) + \eta \cdot x_i(t) \cdot x_j(t)$$

Here, η is the learning rate and x_i, x_j represent activity. When two connected units are active together, their connection becomes stronger. A more complete biological or machine-learning model can include decay, normalization, inhibition, thresholds, or separate plasticity variables. SynapseLab intentionally uses a small transparent model so the learner can see the state changing immediately.

The important point is not that this toy equation reproduces a biological synapse exactly. It is a visible computational abstraction: **recent co-activation changes future behavior by changing connection state**.

4. The SynapseLab Experiment

The interactive experiment starts with an association such as **CAT** → **MEOW**. The learner can then introduce another association, **CAT** → **BARK**, and change the learning process. The interface displays the relative memory strengths so the learner can observe interference rather than treating memory as an invisible

concept.

The experiment follows a simple loop:

① Teach an association → ② update synaptic-memory strength → ③ introduce conflicting information → ④ observe which association becomes stronger → ⑤ explain the result.

This makes the central claim testable. If repeated conflicting input changes the relative strengths, the learner can see interference directly. The demonstration is an educational toy model, not a claim that the displayed numbers are measurements from the biological brain or from an official BDH checkpoint.

5. From Synapses to Dragon Hatchling (BDH)

The connection to **Dragon Hatchling (BDH)** is substantive. The 2025 BDH paper describes a biologically inspired architecture based on locally interacting neuron particles and synapses. During inference, BDH's working memory relies on synaptic plasticity with Hebbian learning using spiking neurons. The authors also report that particular synapses can strengthen when the model processes particular concepts. [3]

This is the key bridge used in SynapseLab: a concept that is introduced as a simple learning experiment also appears as a mechanism in a modern post-Transformer architecture. Instead of treating memory only as a hidden vector, BDH gives synaptic state an explicit computational role.

The BDH section of SynapseLab is deliberately labelled as an **educational/toy illustration**. It is not an official reproduction of BDH and does not claim to run the full published model. This distinction follows the DataForge requirement that independent toy models and reimplementations must be clearly identified. [4]

6. What Changes Compared with a Transformer?

A Transformer primarily uses attention to route information between representations in a sequence, while its learned parameters normally remain fixed during inference. BDH explores a different organization: local neuron-synapse dynamics can make attention-like computation and in-context state emerge through changing synaptic state. The published BDH work presents this as a bridge between Transformer-style sequence learning and brain-inspired local dynamics. [3]

This does not mean that synaptic memory automatically replaces Transformers. Transformers remain highly practical and well studied. BDH itself is a research architecture with open questions about scaling, biological interpretation and broader evaluation. The correct lesson is therefore a design comparison, not a claim of universal superiority.

7. Benefits and Limitations

Potential benefits: temporary information can be represented in a state that changes during inference; the state can be associated with individual connections; sparse/local updates may offer an interpretable micro-level view of computation; and short-term plasticity can improve robustness in some recurrent working-memory settings. [1][3]

Limitations: a toy Hebbian rule is not a complete biological model; interference is an inherent trade-off when multiple associations compete for limited temporary state; and BDH's brain correspondence is a constructive hypothesis rather than proof that biological brains operate exactly this way. The BDH paper itself notes that the chosen graph kernel is not unique and that larger-scale experiments and further ablations remain open questions. [3]

8. Learning Outcome

After using SynapseLab, a learner should be able to explain the difference between fast activity and temporary synaptic state, describe how a Hebbian-style update can create an association, reproduce

interference by introducing a conflicting association, and identify where the same conceptual mechanism appears in BDH. The goal is not to memorize terminology; it is to make the mechanism observable.

9. Evidence and Reproducibility

The experiment in SynapseLab is a transparent educational simulation. The research claims are separated from the simulation and grounded in published work. The artifact exposes the important variable—memory strength—and allows the learner to change the input rather than only watch a scripted animation. This follows the DataForge design principles of one claim, visible state, meaningful controls, fast feedback, and honest labelling. [4]

10. References

- [1] Kozachkov, L., Tauber, J., Lundqvist, M., Brincat, S. L., Slotine, J.-J., & Miller, E. K. (2022). *Robust and brain-like working memory through short-term synaptic plasticity*. PLOS Computational Biology, 18(12), e1010776. DOI: 10.1371/journal.pcbi.1010776.
- [2] Lansner, A., Fiebig, F., & Herman, P. (2023). *Fast Hebbian plasticity and working memory*. Current Opinion in Neurobiology, 83, 102809. DOI: 10.1016/j.conb.2023.102809.
- [3] Kosowski, A., Uznański, P., Chorowski, J., Stamirowska, Z., & Bartoszkiewicz, M. (2025). *The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain*. arXiv:2509.26507.
- [4] DataForge 2026 Pathway Track Problem Statement, supplied for this submission. Requirements include the one-sentence claim, interactive substrate, substantive BDH module, evidence discipline, and clear toy/precomputed labelling.